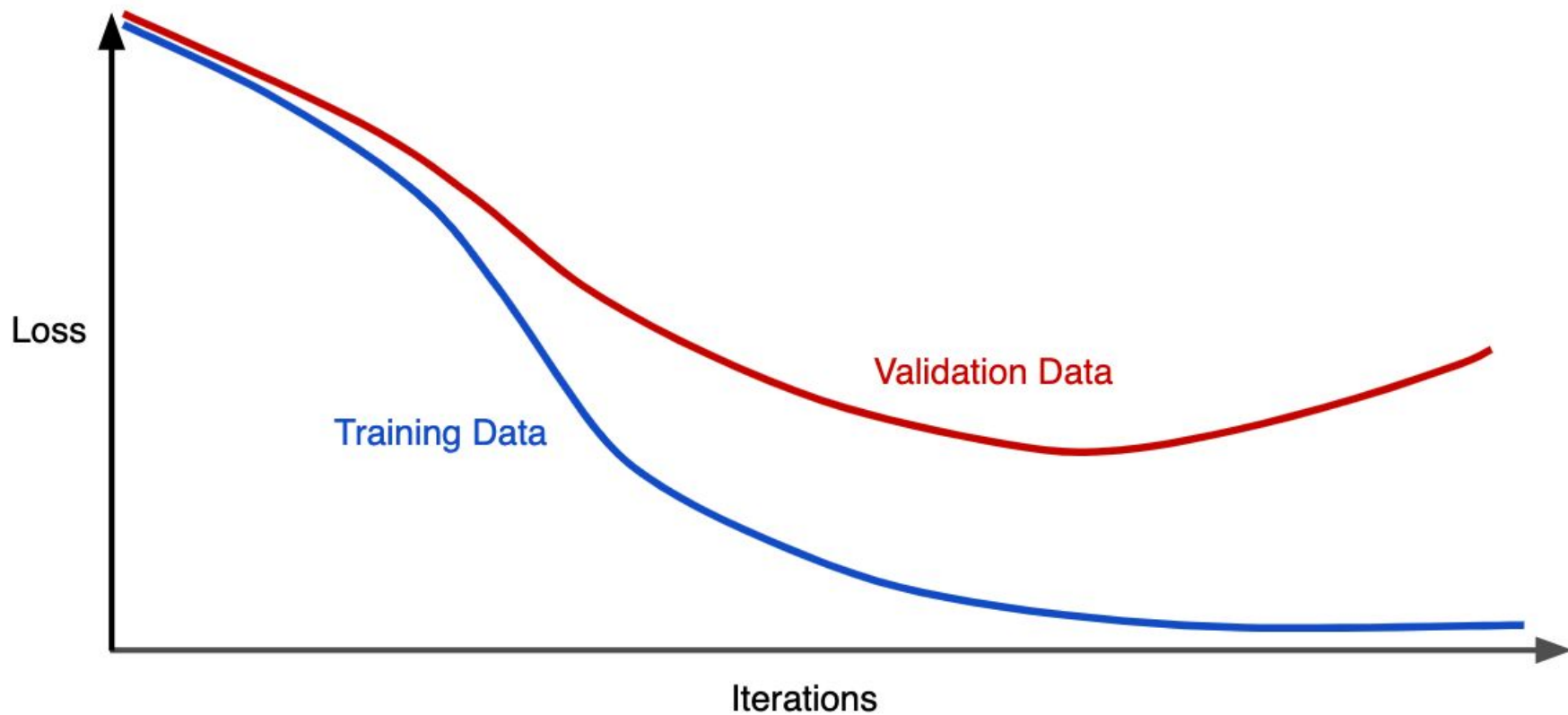


ESE 542 Recitation July 9th, 2021
Regularization and Model Selection

Why regularize?



Penalizing Model Complexity

- We want to avoid model complexity if it is possible
- We try to do this during training when we do optimization
- Our goal:
 - Aim for low training error: $Loss(Data|Model)$
 - Balance against complexity: $complexity(Model)$
- -> minimize: $Loss(Data|Model) + complexity(Model)$

Regularization

- How to define complexity(Model)?
- Prefer smaller weights
- Diverging from this should incur a cost
- Can encode this idea via L_2 regularization (a.k.a. ridge)
 - a. $\text{complexity}(\text{model}) = \text{sum of the squares of the weights}$
 - b. Penalizes really big weights
 - c. For linear models: prefers flatter slopes
 - d. Bayesian prior:
 - i. weights should be centered around zero
 - ii. weights should be normally distribute

A Loss Function with L_2 Regularization

$$Loss(Data|Model) + \lambda(w_1^2 + \dots + w_n^2)$$

Where:

Loss: Aims for low training error

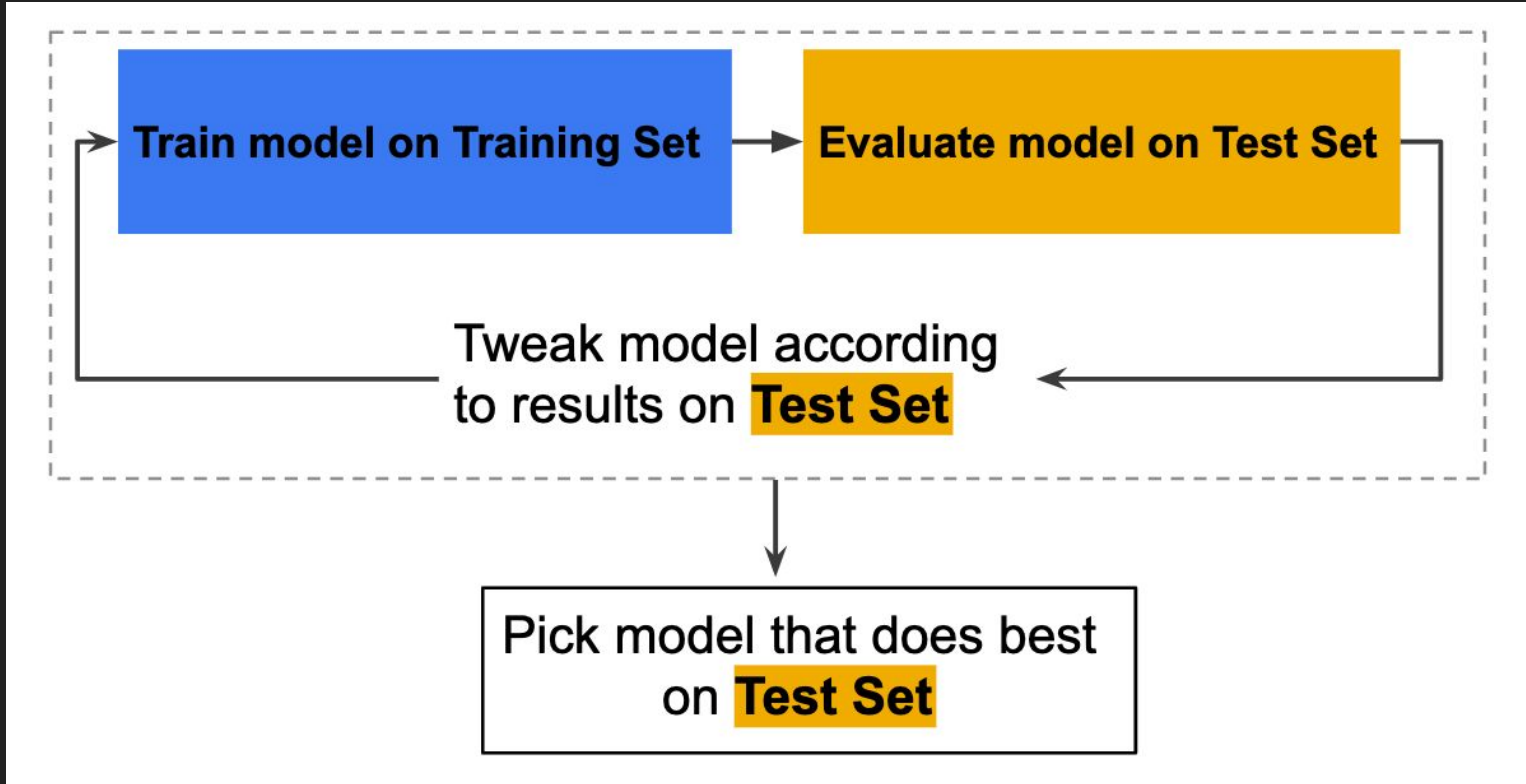
λ : scalar value that controls how weights are balanced

Square of L_2 norm: $w_1^2 + \dots + w_n^2$

Regularization: Sparsity

- Sparse feature crosses may significantly increase feature space
- Possible issues:
 - Model size (RAM) may become huge
 - "Noise" coefficients (causes overfitting)
- L_1 Regularization:
 - Would like to penalize L_0 norm of weights
 - i. Non-convex optimization; NP-hard
 - Relax to L_1 regularization:
 - i. Penalize sum of abs(weights)
 - ii. Convex problem
 - iii. Encourage sparsity unlike L_2

A possible workflow



Partitioning datasets



Better workflow: use validation set

